FIITJEE

ALL INDIA TEST SERIES

FULL TEST – II

JEE (Advanced)-2020

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

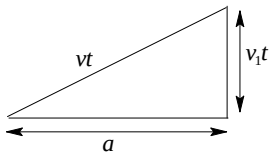
1. A

Sol. $d = 2 + (0.01 \times 67) + (0.01 \times 4)\, mm$

$d = 2.71 mm$

2. D

Sol. $a^2 + v_1^2 t^2 = v^2 t^2$



$t = \dfrac{a}{\sqrt{v^2 - v_1^2}}$

3. C

Sol. Tension in the string will always be zero.

4. B

Sol. $F - F\cos\phi = Ma$

$a = \dfrac{F(1-\cos\phi)}{a}$ towards left.

5. AC

Sol. Wedge will not move since net horizontal force on the wedge is zero.
For block B $T = ma$
$mg - T = ma \Rightarrow \quad a = g/2$

6. ABC
Sol. Friction on block and it's velocity are opposite, hence work is negative.
Friction on plank and its displacement are in same direction, hence work is positive.

7. ACD
Sol. P = weight of the vertical part + force required to provide momentum to the part which is being lefted per unit time
Hence,
$$P = \delta g x + \frac{(\delta v dt)v}{dt}$$
$$P = \delta\left[gx + v^2\right]$$

8. ABD

9. ACD
Sol. Due to symmetrically placed cavities field at center will be zero and potential at equidistant points from center on y-z plane would be equal.

10. ACD
Sol. Buoyant force should be towards point θ.

11. BC
Sol. $v^2 = a - bx$
Differentiating w.r.t. x
$$\frac{2vdv}{dx} = -2bx$$
$$\frac{d^2x}{dt^2} = -bx \Rightarrow motion\, is\, SHM$$
Comparing with
$$\frac{d^2x}{dt^2} = -\omega^2 x$$
$$f = \frac{\sqrt{b}}{2\prod}$$

12. ABC

Sol. Frequency observed by wall $f' = \nu\frac{c+V}{C} \qquad (moving\, observee)$
$=$number of pulse striking persecond

Frequency of reflected wave $f'' = f'\frac{c}{C-V} = \nu\frac{C+V}{C-V}$

Wavelength $f'' = f'\frac{c}{C-V} = \nu\frac{C+V}{C-V}$

SECTION – C

13. 00074.50

Sol. Pressure of air above Hg column

$P_1 = P_0 - 70 \qquad cm of\ Hg$

$P_2 = P_0 - 67$

$P_2 = P_1 \times \frac{10}{6}$

$\frac{10}{6} P_1 = P_0 - 67$

$P_1 = \frac{6P_0}{10} - 40.2$

$P_0 - 70 = \frac{6P_0}{10} - 40.2$

$\frac{4P_0}{10} = 29.8$

$P_0 = \frac{29.8 \times 10}{4} = 74.5 cm$

14. 01260.00

Sol. Potential difference without introducing the slab is given by

$V_0 = E \times d = 300 \times 5 = 1500$ volt

The capacitance C_0 is given by

$C_0 = \frac{\in_0 A}{d} = \frac{\in_0 A}{0.05}$ farad

$\therefore \quad \in_0 A = 0.05\ C_0 \qquad ...(1)$

when the slab is inserted, the capacitance C is given by

$C = \frac{\in_0 A}{d - t + t/K}$ where t = thickness of slab

$= \frac{0.05 C_0}{0.04 + \frac{0.01}{5}} = \frac{25 C_0}{21} \qquad ...(2)$

Now $C_0 V_0 = CV$

$C_0 \times 1500 = \frac{25 C_0}{21} \times V$

or $V = \frac{1500 \times 21}{25} = 1260$ volt

15. 00010.00

Sol. Area of the piston = $\pi (R^2 - r^2)$

Pressure on the base = $\rho g (H + h)$ where

ρ = density of water.

Also pressure on the base

$= \frac{(W + m)g}{\pi R^2}$

$\therefore \rho g (H + h) = \dfrac{(W + m)g}{\pi R^2}$

$\Rightarrow$ H + h $\dfrac{(W + m)}{\pi R^2 \rho}$ .....(i)

Considering the equilibrium of the piston
Upward thrust = weight of the piston
$\pi(R^2 - r^2)$ □gh = Wg

$\Rightarrow$ h = $\dfrac{W}{\pi(R^2 - r^2)\rho}$ .....(ii)

From (i) and (ii)

H = $\dfrac{W + m}{\pi R^2 \rho} - \dfrac{W}{\pi(R^2 - r^2)\rho}$

$\Rightarrow$ H = $\dfrac{1}{\pi R^2 \rho}\left(m - W\dfrac{r^2}{R^2 - r^2}\right)$

$\Rightarrow$ H = $\dfrac{1}{\pi \times 4^2 \times 10^{-4} \times 1000}$

= $\left(0.7 - \dfrac{3 \times 1^2 \times 10^{-4}}{(4^2 - 1^2) \times 10^{-4}}\right)$

= 10 cm

16. 00006.00

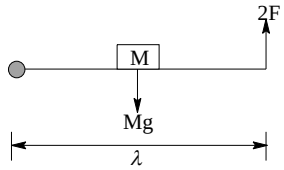
Sol. By linear momentum conservation impulse (J) = mV.

By angular momentum conservation, angular impulse $= J\dfrac{\lambda}{2} = I\omega$

So $mv\dfrac{\lambda}{2} = I\omega \; or \; \omega = \dfrac{mv\lambda}{2I} = \dfrac{mv\lambda}{2\left(\dfrac{m\lambda^2}{12}\right)} = \dfrac{6v}{\lambda} = 6\,rad/s$

17. 00005.00

Sol.



For equilibrium

$\left|Mg\dfrac{\lambda}{2}\right| = |2F|1$

$F = \dfrac{mg}{4} = \dfrac{100}{4} = 25\,N$

18. 00003.00

Sol. From Wien's displacement law and Stefen Boltzman law the wavelength of the peak radiation is halved.

So, $\lambda = 3000\mathring{A}$

$$\frac{hc}{\lambda} - \frac{\alpha}{100} = 13.6\left[\frac{1}{4} - \frac{1}{16}\right]$$

$$\frac{hc}{\lambda} - \frac{\alpha}{100} = 2.55$$

$$\frac{12420}{3000} - \frac{\alpha}{100} = 2.55$$

$$\frac{\alpha}{100} = 4.14 - 2.55 = 1.59$$

$$\alpha = 159 = 53x$$

$$x = 3$$

Chemistry

PART – II

SECTION – A

19. C
Sol. Fully filled d-subshell is colourless in nature.

20. D

Sol. $\Delta S = RLn\frac{V_2}{V_1}$

$\Delta H = 0$

$\Delta G = \Delta H - T\Delta S$

21. A
Sol. Salt bridge does not participate in chemical reaction.

22. C

23. CD

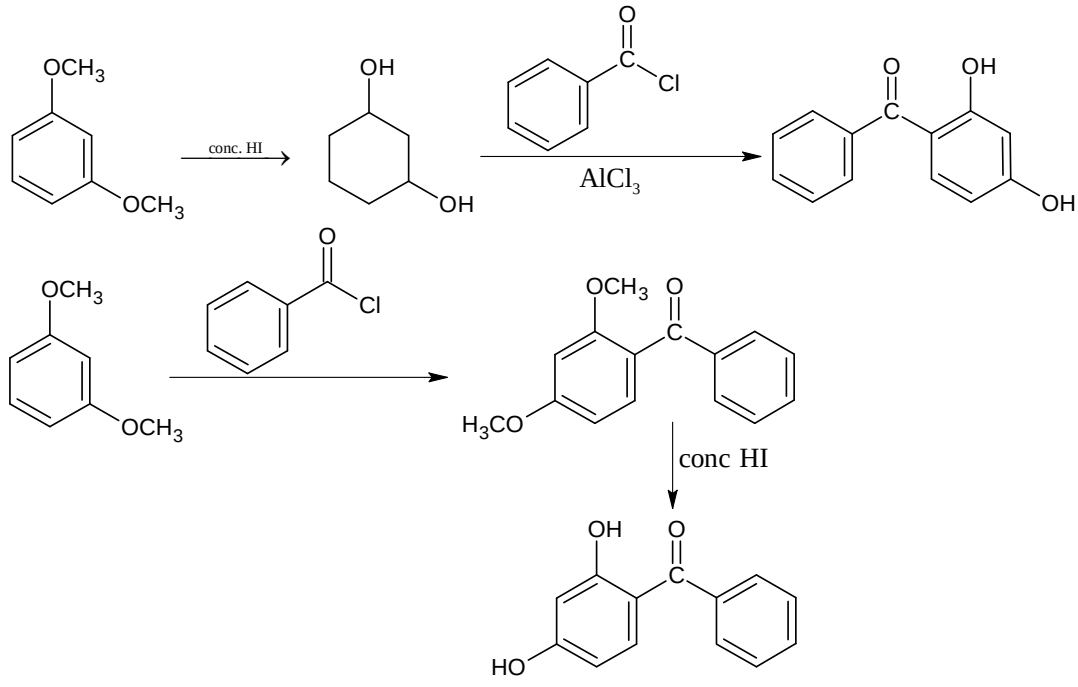
24. AB
Sol. Solute X will dissociate into the solution so i>1.

25. ABD
Sol. $\left[CoF_6\right]^{3-}$ is a high spin complex.

26. ABCD

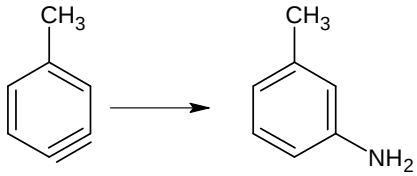
27. ACD
Sol. Due to the more cohesive energy in Li, it has highest melting point among alkali metals.

28. BC
Sol.



29. AB

Sol.



Intermediate product

30. AC

Sol. This is β-D- fructofuranose and a reducing sugar.

SECTION – C

31. 00500.00

32. 00100.00

Sol. $K = Ae^{-Ea/RT}$

33. 09900.00

Sol. $\Delta U = q\ tw$

For adiabatic process $q = 0$

$\Delta U = w$

$\Delta U = P\Delta V = -P\left(V_2 - V_1\right)$

$\Delta U = -100\left(99 - 100\right) = 100\,\text{bar mL}$

Now $\Delta H = \Delta U + \Delta\left(PV\right)$

$\Delta H = 100 + \left[100 \times 99 - 1 \times 100\right]$

$\Delta H = 9900\,\text{bar mL}$

34. 00098.00

Sol. $T = 20°C$ $0.1m$ solution of urea ΔP=0.03mm

$KCl \rightarrow 0.1m$ $\Delta P = 0.0594$

$\frac{\Delta P}{P} = x_\beta \rightarrow 0.1 mole\ in\ 1kg$

$\frac{0.03}{P} = \frac{0.1}{0.1 + 55.5}$

$P = \frac{0.03 \times 55.65}{0.1} = 16.695$

$\frac{\Delta P}{P} = iX_\beta$

$\frac{0.0594}{16.695} = i\left[0.00179\right]$

$i = 1.987$

35. 00036.00

36. 00030.60

Sol. $KE = 13.6 \times \frac{Z^2}{n^2} = 13.6 \times \frac{3^2}{2^2} = 30.6\text{eV}$

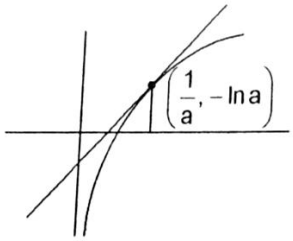
Mathematics

PART – III

SECTION – A

37. A

Sol.



$f(x) = \ln x$

$f'(x) = \frac{1}{x}$

$\Rightarrow x = \frac{1}{a}$ at point of tangency

$\frac{-\ln a + 1}{1 + \frac{1}{a}} = a \Rightarrow a + \ln a = 0$

Let $f(a) = a + \ln a = 0$

$f\left(\frac{1}{e}\right) < 0 \; and \; f\left(\frac{1}{\sqrt{e}}\right) > 0$

$\Rightarrow a \in \left(\frac{1}{e}, \frac{1}{\sqrt{e}}\right)$

38. C

Sol. $|z_2 + i z_1| = |z_1| + |z_2|$

$\Rightarrow z_2, i z_1, 0$ are collinear

$\Rightarrow \arg(i z_1) = \arg z_2$

$z_3 = \frac{z_2 - i z_1}{1 - i} \Rightarrow (1 - i) z_3 = z_2 - i z_1$

$\Rightarrow i = \frac{z_3 - z_2}{z_3 - z_1} \Rightarrow \arg\left(\frac{z_3 - z_2}{z_3 - z_1}\right) = \frac{\pi}{2}$

$and \; |z_3 - z_2| = |z_3 - z_1|$

$\Rightarrow AC = BC, AB^2 = AC^2 + BC^2$

$\Rightarrow AC = \frac{5}{\sqrt{2}}; \Delta = \frac{1}{2} \times \frac{5}{\sqrt{2}} \times \frac{5}{\sqrt{2}} = \frac{25}{4} \; sq.units$

39. A

Sol. $\left({}^3C_1\right)^5$

40. D

Sol. $f(t)=g(t)\Rightarrow f'(t)=-f(t)\Rightarrow=0$

$\because \{f(t)\text{is one}-\text{one function}\}$

$t=0, x=y=1$

$\because \frac{dy}{dx}=-\frac{1}{x^2} and \frac{d^2y}{dx^2}=\frac{2}{x^3}$

$at\ x=1, \frac{d^2y}{dx^2}=2$

41. ACD

Sol: $\because (A+B)(A-B)=(A-B)(A+B)$

Now $C=\left(A^T+B^T\right)(A+B)(A-B)^{-1}$

$=(A-B)(A+B)(A-B)^{-1}$

$=(A+B)(A-B)(A-B)^{-1}$

$=A+B$

Now $C^T=A^T+B^T=A-B$

$C+C^T=2A$

42. BD

Sol: Let $t_r=\left(\frac{1}{r(r+1)}\right)^3=\left(\frac{1}{r}-\frac{1}{r+1}\right)^3=\frac{1}{r^3}-\frac{1}{(r+1)^3}-\frac{3}{r(r+1)}\left(\frac{1}{r(r+1)}\right)$

$=\frac{1}{r^3}-\frac{1}{(r+1)^3}-3\left(\frac{(r+1)-r}{r(r+1)}\right)^2$

$t_r=\frac{1}{r^3}-\frac{1}{(r+1)^3}-3\left(\frac{1}{r^2}+\frac{1}{(r+1)^2}\right)+6\left(\frac{1}{r}-\frac{1}{r+1}\right)$

$\sum_{r=1}^{n}t_r=\left(1-\frac{1}{(n+1)^3}\right)-3\left(\frac{\pi^2}{6}+\frac{\pi^2}{6}-1\right)+6\left(1-\frac{1}{n+1}\right)$

$\lim_{x\to\infty}\sum t_r=10-\pi^2$

43. BCD

Sol. $f(x)=x^2+\int_0^x e^{-t}f(x-t)\,dt$

$= x^2 + \int_0^x e^{-(x-t)}.f\left(x(x-t)\right)dt$

$= x^2 + e^{-x}.\int_0^x e^t.f\left(t\right)dt$

$\therefore e^x.f\left(x\right) = x^2.e^x + \int_0^x e^t.f\left(t\right)dt$

Differentiate with respect to x, we get $f'(x) = x^2 + 2x$

$\Rightarrow f\left(x\right) = \frac{x^3}{3} + x^2 + x, also\ f\left(0\right) = 0$

$\Rightarrow c = 0$

$\Rightarrow f\left(x\right) = \frac{x^3}{3} + x^2$

44. AC

Sol. Obvious

45. ABC

Sol. When 9 is not included $= \frac{{}^8C_3.{}^8C_3 - {}^8C_3}{2}$

and required probability is $= \frac{1}{3} + = \frac{{}^8C_3.{}^8C_3 - {}^8C_3}{2.{}^9C_3.{}^8C_3} = \frac{37}{56}$

46. AB

Sol. By property: $PQ \cdot P'Q = b^2$ and area of $\Delta CLG = ab$

47. ABC

Sol. $0 < x < \frac{\pi}{2} \Rightarrow \frac{\sin x}{x}$ is decreasing and $\sin x < x < \tan x$

$\Rightarrow \frac{\sin\left(\sin x\right)}{\sin x} > \frac{\sin x}{x} > \frac{\sin\left(\tan x\right)}{\tan x}$

$\Rightarrow I_1 > I_2 > I_3$

48. ABC

Sol. a, b, c in $A.P.$ so line concurrent at $(1, -2)$

$\therefore$ Centre of auxiliary circle $\left(-\alpha, -\beta\right) = (1, -2)$ radius of circle $= 2$

Length of major axis $= 4 = 2A$

$\because \frac{1}{SP} + \frac{1}{SQ} = \frac{2A}{B^2}$

$\Rightarrow B = \sqrt{3}$

Hence, $e = \frac{1}{2}$

SECTION – C

49. 00005.00

Sol. Put $z=0$ in the given equation $x=5 \text{ and } y=1$

$\Rightarrow 5.1=c^2$

50. 00009.00

Sol. Let $\sqrt{3a+5}=x, \sqrt{3b+5}=y, \sqrt{3c+5}=z$

$\left(x+y+z\right)^2 \le 3\sum x^2; \left(\because \sum x^2 \ge \sum xy\right)$

51. 00008.00

Sol. Let $f\left(x\right)=\dfrac{x}{70+x^2}; f'\left(x\right)=\dfrac{70-x^2}{\left(70+x^2\right)^2}$

So $f\left(x\right)$ increase in $\left(0,\sqrt{70}\right) \text{ and } f\left(x\right)$ decrease in $\left(\sqrt{70},\infty\right)$

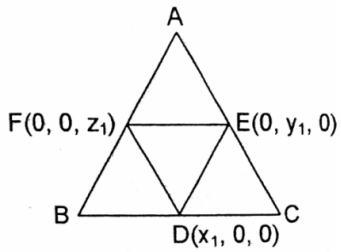
So, greater term may be $a_8 \text{ or } a_9$

But $\dfrac{a_8}{a_9}>1$

$\Rightarrow k=8$

52. 00008.00

Sol.



Let the intercept of plane on the axes be

$\left(x_1,0,0\right),\left(0,y_1,0\right),\left(0,0,z_1\right)$

So, equation is $\dfrac{x}{x_1}+\dfrac{y}{y_1}+\dfrac{z}{z_1}=1$

$\because EF \parallel BC \text{ and } EF=\dfrac{1}{2}BC=4\left(y_1^2+z_1^2\right)=a^2$

Similarly $4\left(x_1^2+y_1^2\right)=c^2 \text{ and } 4\left(z_1^2+x_1^2\right)=b^2$

$\Rightarrow a^2+b^2+c^2=8\left(x_1^2+y_1^2+z_1^2\right)$

$\Rightarrow k=8$

53. 00001.00

Sol. $\tan\left(2-1\right)=\dfrac{\tan 2-\tan 1}{1+\tan 2\cdot\tan 1}$

$$\Rightarrow \tan 1^\circ \tan 2^\circ = \frac{\tan 2 - \tan 1}{\tan 1} - 1 \text{(Similarly)}$$

54. 00008.00

Sol. $t_1 + t_2 = 2$

$$\frac{2 - m_2}{1 + 2m_2} = \sqrt{3}$$

$$\text{and } \frac{m_3 - 2}{1 + 2m_3} = \sqrt{3}$$

$$t_1 + t_2 + t_3 = \frac{p}{q} = \frac{3}{11}$$